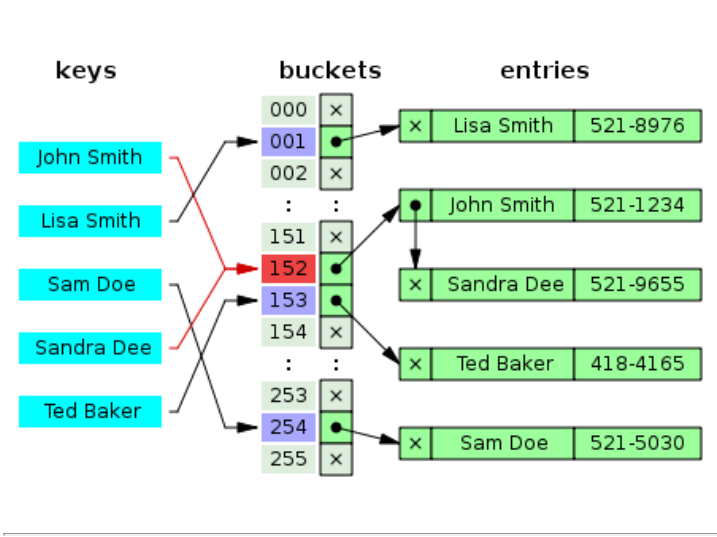
Consistent Hashing

Hash Tables

- Goal: Efficiently store and search data
- Naive approach: Array or linked list
  - Search by iterating elements → **O(n)** complexity
  - Fast for small lists, slow for large datasets
- Hash Functions:
  - Map arbitrary data → integer index
  - Allows near-constant time access for non-sequential or random keys
  - Formula: `index = hash(object) mod N`
    - N = array size (practical, usually ~2× expected elements)

Handling Collisions

- Multiple objects can map to the same index → **collisions**
- Typical solution: **buckets**
  - Each array entry points to a list of objects
  - Insert: add to bucket if not present
  - Search: check bucket for object
- Complexity: ~**O(N/k)**, with k = number of buckets


keys
buckets
entries
John Smith
Lisa Smith
Sam Doe
Sandra Dee
Ted Baker
000
001
002
151
152
153
154
253
254
255
Lisa Smith
521-8976
John Smith
521-1234
Sandra Dee
521-9655
Ted Baker
418-4165
Sam Doe
521-5030

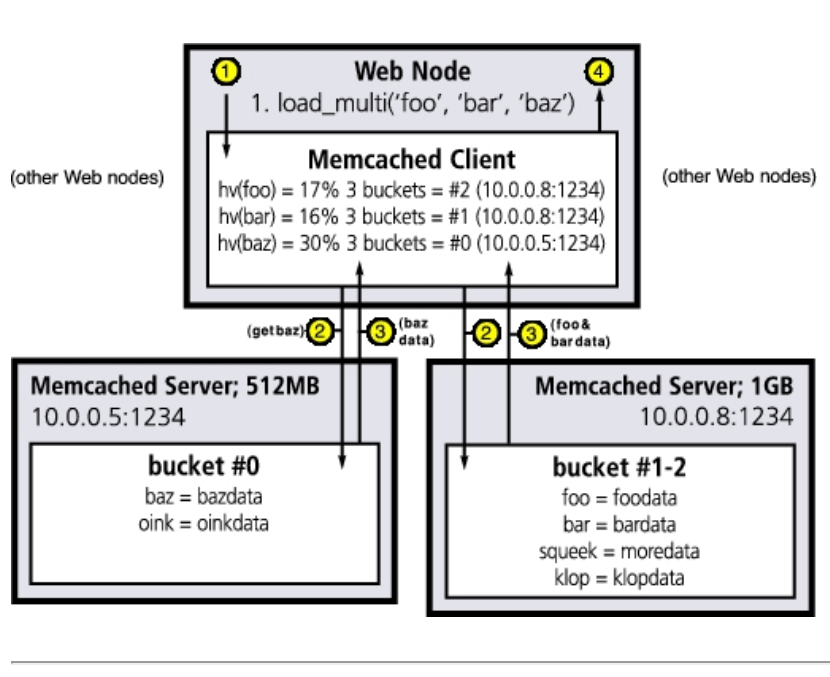
Distributed Hash Tables & Key Distribution

- **Motivation:**
  - Split a hash table across multiple servers to bypass memory limits
  - Enable arbitrarily large hash tables
- **How It Works:**
  - Objects and keys are distributed among several servers → **Distributed Hash Table (DHT)**
  - Typical use case: **in-memory caches** (e.g., **Memcached**)
    - Pool of servers stores key/value pairs
    - Provides fast access to frequently used data

Cache Workflow

1. Application queries the cache first
2. If **cache miss**, fetch from database/source
3. Store result in cache for future access

- **Key Distribution:**
  - Strategy determines **which keys go to which servers**
  - Crucial for **load balancing, scalability, and reliability**


Web Node
1. load_multi('foo', 'bar', 'baz')
Memcached Client
hv(foo) = 17% 3 buckets = #2 (10.0.0.8:1234)
hv(bar) = 16% 3 buckets = #1 (10.0.0.8:1234)
hv(baz) = 30% 3 buckets = #0 (10.0.0.5:1234)
(other Web nodes)
(other Web nodes)
(get baz)
(baz data)
(foo & bar data)
Memcached Server; 512MB
10.0.0.5:1234
bucket #0
baz = bazdata
oink = oinkdata
Memcached Server; 1GB
10.0.0.8:1234
bucket #1-2
foo = foodata
bar = bardata
squeek = moredata
klop = klopdata

Key Distribution via Hash Modulo

- Formula:

```
server_index = hash(key) mod N
```

  - N = number of servers
  - Client computes hash → modulo → contacts server
  - Same hash function across clients is required

Example Setup

- Servers: **A, B, C** (N = 3)
- Keys and their hashes:

KEY	HASH	HASH mod 3
john	1633428562	2
bill	7594634739	0
jane	5000799124	1
steve	9787173343	0
kate	3421657995	2

A	B	C
bill	jane	john
steve		kate

The Rehashing Problem

- Simple modulo-based distribution works **until the number of servers changes**
- Problem occurs when:
  - A server crashes or becomes unavailable
  - New servers are added to the pool
- Most keys need to be **rehased** because `hash(key) mod N` changes

Example: Removing Server C

- Previous pool: 3 servers (A, B, C)
- Remove server C → N = 2
- Recompute server index: `server_index = hash(key) mod 2`

KEY	HASH	HASH mod 2
john	1633428562	0
bill	7594634739	1
jane	5000799124	0
steve	9787173343	1
kate	3421657995	1

A	B
john	bill
jane	steve
	kate

- **Observation:** All key locations changed, not just those on removed server

---

Consequences for Caching

- Most queries result in **cache misses**
- Data must be fetched again from the source
- Heavy load on origin servers (e.g., databases)
- Can severely **degrade performance** or cause **server crashes**

The Solution: Consistent Hashing

- Problem: Modulo-based distribution requires massive rehashing when servers change
- Solution: **Consistent Hashing** (Karger et al., 1997)
- Minimizes key relocation when servers are added or removed

Hash Ring Concept

- Map the hash output range to a circle:
  - 0 → 0°
  - INT_MAX → 360°
  - All other hashes → intermediate angles
- Keys and servers are placed on the same circle by hashing their values

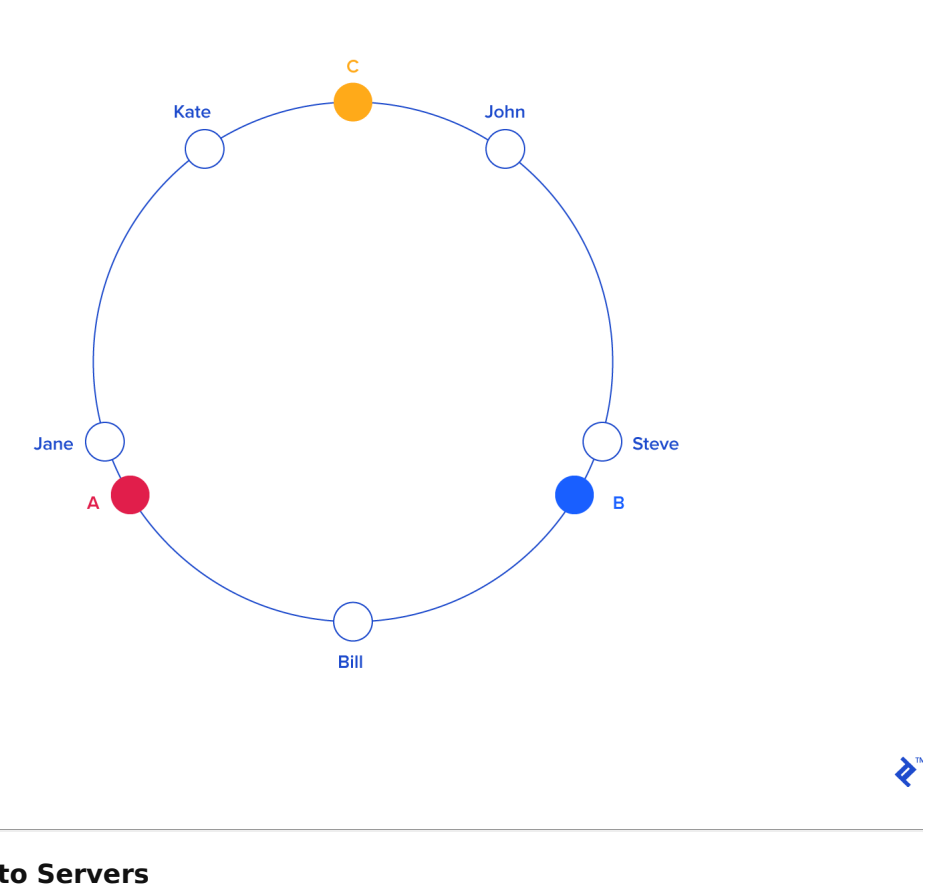
---

Example: Keys on the Circle

KEY	HASH	ANGLE (DEG)
john	1633428562	58.8
bill	7594634739	273.4
jane	5000799124	180
steve	9787173343	352.3
kate	3421657995	123.2

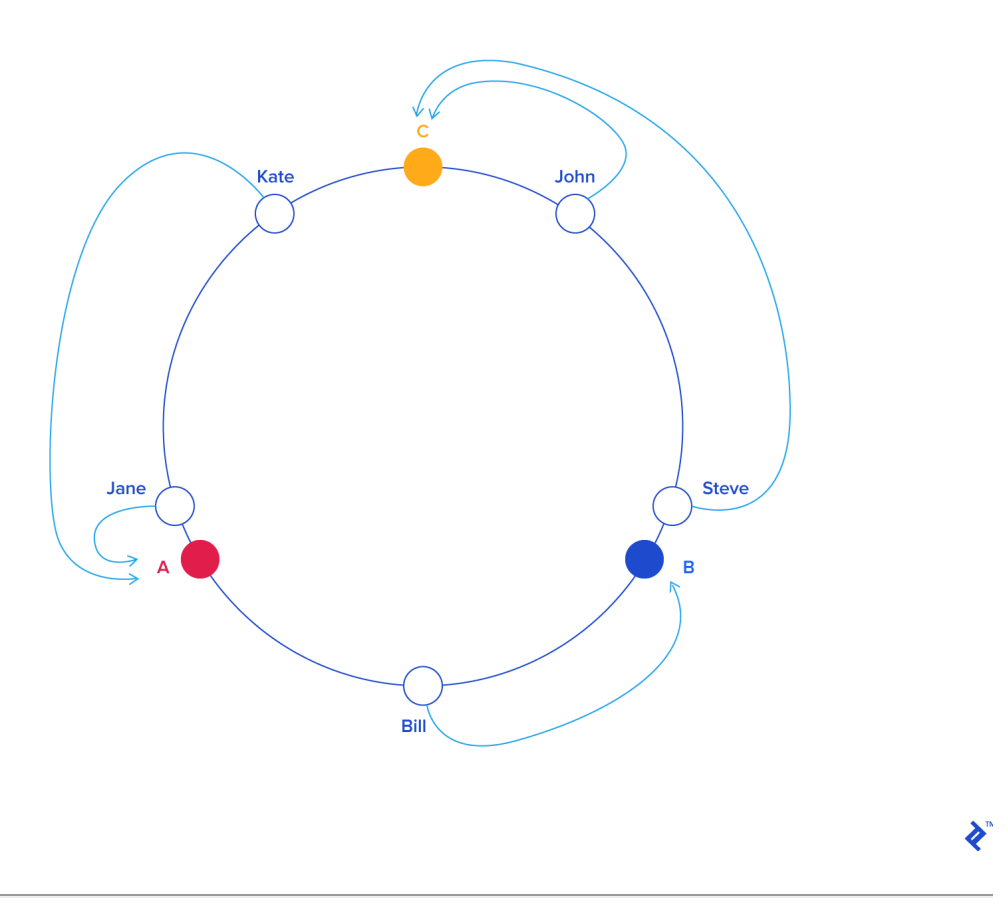
- Servers are also hashed to angles:

SERVER	HASH	ANGLE (DEG)
A	5572014558	200.6
B	8077113362	290.8
C	2269549488	81.7


C
Kate
John
Jane
Steve
A
B
Bill

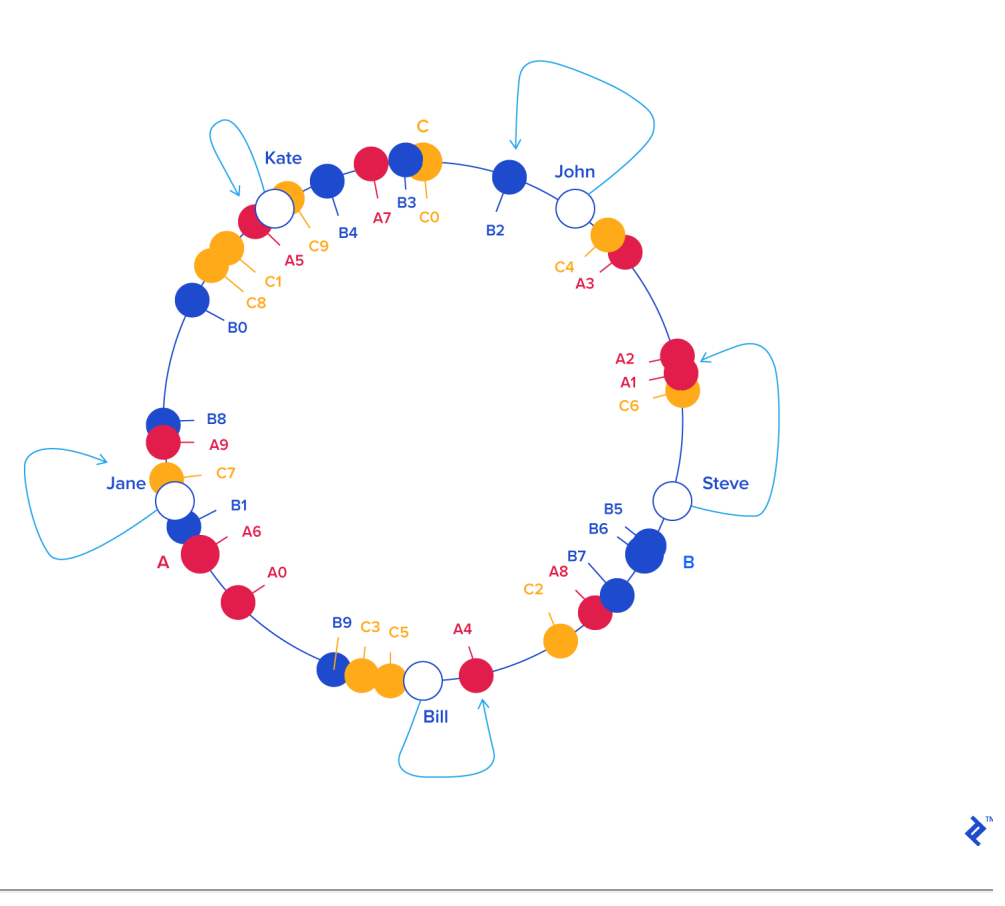
Mapping Keys to Servers

- Rule: **Each key goes to the first server clockwise after its angle**
- Example assignments:


C
Kate
John
Jane
Steve
A
B
Bill

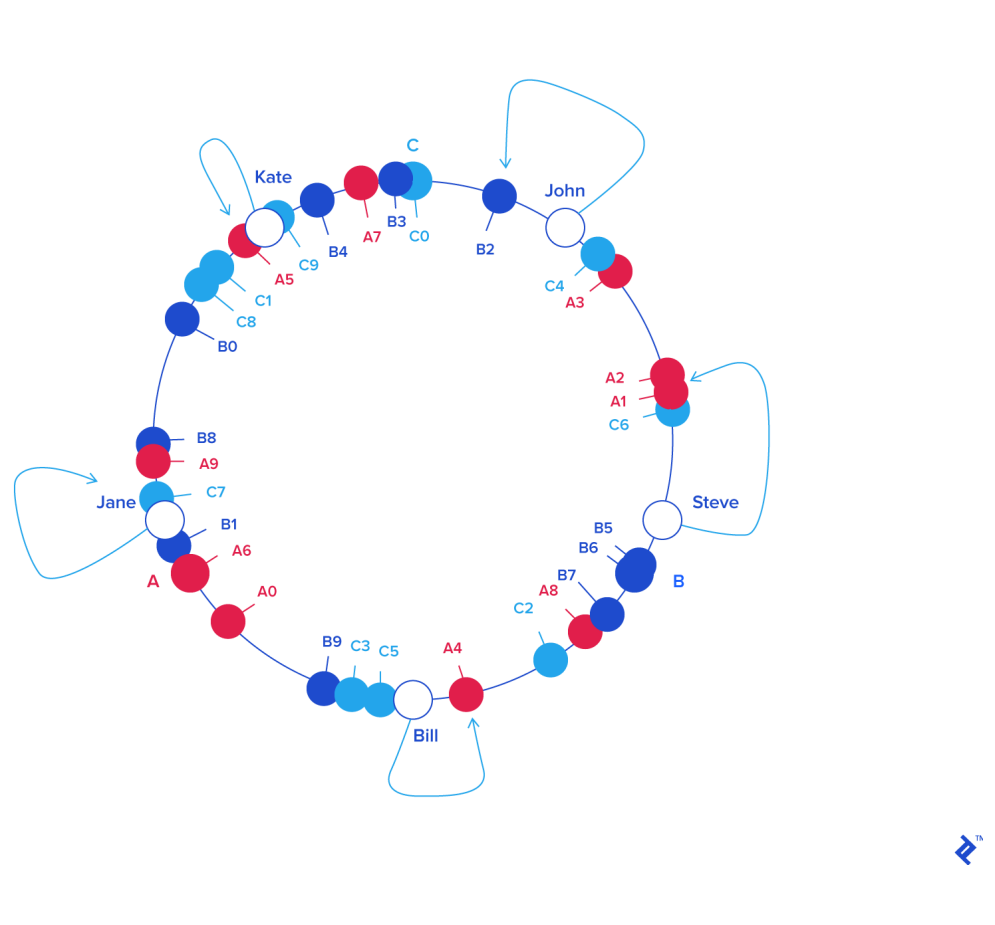
Balancing Load: Virtual Nodes

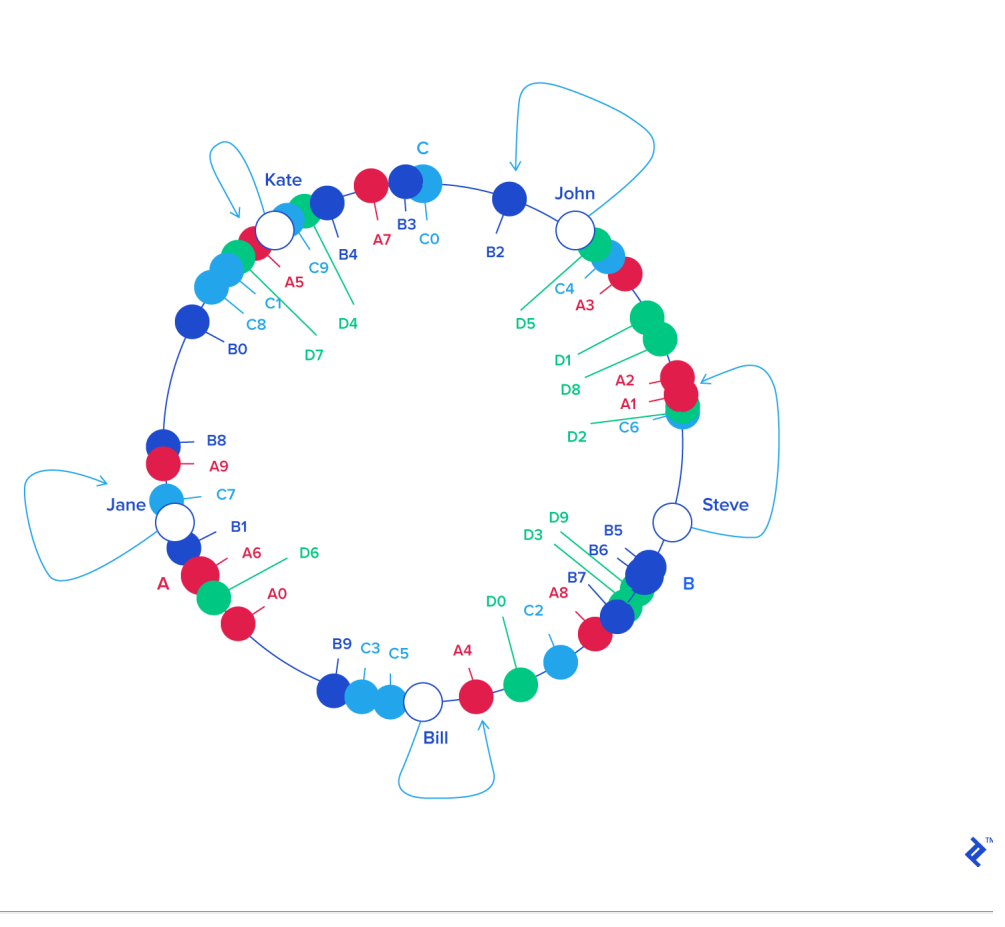
- Assign multiple **labels/angles per server** (virtual nodes)
- Example: A0..A9, B0..B9, C0..C9
- Adjust weight per server for load balancing:
  - More powerful servers → more virtual nodes


C
Kate
John
Jane
Steve
A
B
Bill
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
B0
B1
B2
B3
B4
B5
B6
B7
B8
B9
C0
C1
C2
C3
C4
C5
C6
C7
C8
C9

Resilience to Server Changes

- **Server removal:** only keys mapped to removed server labels are reassigned
- **Server addition:** only keys near new server labels are reassigned
- Keys for other servers **remain unchanged**
- Result: **minimal remapping** → avoids massive cache misses


C
Kate
John
Jane
Steve
A
B
Bill
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
B0
B1
B2
B3
B4
B5
B6
B7
B8
B9
C0
C1
C2
C3
C4
C5
C6
C7
C8
C9


C
Kate
John
Jane
Steve
A
B
Bill
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
B0
B1
B2
B3
B4
B5
B6
B7
B8
B9
C0
C1
C2
C3
C4
C5
C6
C7
C8
C9
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9

Replication Factor

- Store each key on **k different servers**
- Ensures **fault tolerance**: data available if a server fails
- Example: Replication factor = 3 → key `john` on B, C, and A

Resources

- https://www.linuxjournal.com/article/7451
- https://www.toptal.com/developers/big-data/consistent-hashing